PyRAMIDS Manual

A Python package for Radiation, Magnetoelectric Interactions, and Dipoles in Stratified Layers

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Overview

PyRAMIDS is a Python package for simulating electromagnetic radiation, magnetoelectric interactions, and dipole-based multiple scattering in stratified media. This manual briefly summarizes the code architecture and folder structure, lists external dependencies (with tested versions), and points to benchmarks and example scripts. Example cases include literature benchmarks, scripts to reproduce figures in the paper, and internal validation tests used for continuous consistency checking.

Contents

Library Structure 2
Benchmarks 3
Examples and UserSandbox 3

Folder Structure Overview

PyRAMIDS is implemented in *Python* (v3.12.7) and distributed together with benchmark tests and usage examples. The computational core relies on *NumPy* (v1.26.4) and *SciPy* (v1.15.1), with extensive vectorization for numerical efficiency. Performance-critical routines are accelerated using *Numba* JIT compilation (v0.60.0), which can be enabled, disabled, or cached depending on user preference. Cached compilation is recommended for routine usage, as it combines compilation speed-up with minimal overhead during repeated runs. During development and debugging of core routines, disabling JIT is advisable. Visualization routines use *Matplotlib*. A compatible Python environment can be created by installing the pinned dependencies, for example using `pip install package==version`. The versions listed above correspond to the development and testing environment used by the authors; PyRAMIDS should be generally compatible with earlier and more recent releases of these packages, although only the versions specified here have been formally validated.

The package is organized into the following top-level directories:

Library
: Core computational framework implementing the mathematical formalism described in the Supplementary Material.

Benchmarks
: Validation routines, including literature benchmarks and internal consistency tests.

Examples
: Typical usage workflows and scripts to reproduce figures from this chapter.

UserSandbox
: Workspace for user-developed scripts, including batch processing utilities to execute benchmarks and examples.

Library Structure

The **Library** folder is divided into three subdirectories: **Core**, **Utility**, and **Use**. This layered structure separates low-level physics implementations from user-facing interfaces.

Core

The **Core** directory contains low-level implementations written in a slab-centric coordinate system. These routines form the computational backbone of the package. Direct use is discouraged unless the user is extending the framework.

`Core_Smatrix.py`
: Implements the S-matrix formalism for arbitrary stratified multilayer systems. The code constructs interface and propagation matrices and recursively builds the full stack response using the Redheffer star product for both s and p polarizations. Reflection and transmission coefficients in semi-infinite media are computed, along with internal field amplitudes.

`Core_ImGLDOS.py`
: Computes electric, magnetic, and magnetoelectric LDOS using contour integration of the dyadic Green tensor. Adaptive integration is performed via `scipy.integrate.quadvec`.

`Core_Radiationpattern.py`
: Evaluates the far-field radiation pattern of an arbitrary magnetoelectric dipole $(p_x, p_y, p_z, m_x, m_y, m_z)$ embedded in a stratified medium using the asymptotic dyadic Green function. Fields are returned in the s-p basis for specified (θ, ϕ) angles. Integrated radiated power in each half-space is also provided.

`Core_Greenslab.py`
: Computes generalized dyadic Green functions within a layered slab using the angular spectrum representation, assuming emitter and detector reside in the same layer.

Utility

The **Utility** directory provides helper functions that interface between user-level inputs and the Core routines.

`Util_argumentchecker.py`
: Validates user input consistency such as dipole dimensionality, complex values, stack definitions, and many more. Reshapes arrays to match Core expectations.

`Util_argumentwrapper.py`
: Converts user-centric coordinates (first interface at $z = 0$) into slab-centric coordinates for internal computations.

`Util_vectorpolarization.py`
: Provides visualization routines for far-field emission patterns and Stokes polarimetry, enabling Fourier-plane emission analysis analogous to experimental back-focal-plane imaging.

`Util_differential_evolution_custom_constraints.py`
: Provides a custom Differential Evolution optimizer with constraint handling. It is used for parameter optimization and inverse-design routines within the package.

Use

The **Use** directory contains the high-level user interface. These scripts operate in a global (user-centric) coordinate system and internally call Utility and Core modules.

`Use_Planewave.py`
: Computes reflectance, transmittance, absorptance, and layer-resolved absorption for s- and p-polarized plane waves incident at arbitrary angles. Includes field-distribution visualization routines.

`Use_LDOS.py`
: Computes electric, magnetic, and magnetoelectric LDOS for canonical and arbitrary dipole orientations at any position within the stack. Provides access to $k_{\parallel}$-resolved integrands for modal analysis.

`Use_Radiationpattern.py`
: Calculates angle-resolved far-field emission and radiated power. Radiative LDOS can be obtained via angular integration.

`Use_Multiplescattering.py`
: Solves multiple-scattering problems for dipole scatterers in stratified media. The coupling matrix is constructed from inverse polarizabilities and Green-function interactions. The module computes induced dipole moments, extinction cross-sections (via work done), and scattering cross-sections (via far-field integration). Radiation-damping corrections consistent with the optical theorem are included.

Benchmarks

The **Benchmarks** directory contains:

Literature
: Reproduces established results from Refs. [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17].

Internal_Consistency
: Performs cross-validation between independent implementations:

- LDOS from ImG vs radiation-pattern integration
- Green-function LDOS vs ImG LDOS
- Homogeneous-medium Green tensor vs analytical form
- Optical-theorem validation in multiple scattering
- Pseudochiral dipole rotation invariance tests

These checks ensure physical consistency and numerical robustness across the full framework.

Examples and UserSandbox

The **Examples** directory provides scripts demonstrating standard workflows and reproducing figures from this chapter.

The **UserSandbox** directory serves as a customizable workspace for user-defined problems and includes batch-execution scripts for running all benchmark and example files.

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